

MANAS AGARWAL

Ghaziabad, Uttar Pradesh

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EDUCATION

Dr. A.P.J. Abdul Kalam Technical University

Sept 2022 - Sept 2026

Bachelor of Technology in Computer Science and Information Technology (CSIT).

Senior Secondary School, CBSE , 92.2%

2022

Saraf Public School

TECHNICAL SKILLS

Technologies and Frameworks: ReactJS, Redux Toolkit, NodeJS, Express, Sequelize ORM

Databases and Cloud: MongoDB, MySQL, PostgreSQL, Amazon Web Services (AWS), Docker

Languages and Tools: SQL, TypeScript, JavaScript, C, C++ , Visual Studio Code, Git, Github, Postman, etc.

EXPERIENCE

Coordinator, Programming Club

Feb 2024 – Present

- Developed a scalable Programming Club website with a team, following industry best practices. **Programming Club**
- Worked and managed a team of **40+ members** to conduct various technical activities, doubt sessions, and coding competitions.
- Conducted a series of Data Structure and Algorithm workshops, named **#include** and mentored over 100+ students.

PROJECTS

Chess Master | *WebRTC, Socket.io, Sequelize ORM, PostgreSQL* | 🌐 [Github](#) | 🎥 [Video Demo](#) | 📄 [Site Link](#)

- Leveraged **WebRTC** to let players engage in **live discussions** during gameplay, enabling them to share strategies, learn tricks, and recognize patterns for mastering chess, **serving over 100+ players**.
- Empowers players to **revisit and analyze their past games**, providing insights into their best moves and misjudgments to enhance their performance.
- Developed a adaptive **rating system** starting at 1200, incorporating **mathematical and logical calculations** to accurately reflect player skill levels up to 2000.
- **Allow players to challenge their friends** and also ensures a secure sign-in and sign-up process through **OAuth integration**.

VStream | *ReactJS, Redux Toolkit, NodeJS, Express, MongoDB* | 🌐 [Frontend](#) | 🌐 [Backend](#) | 📄 [Site Link](#)

- VStream is a **unified video and music streaming** platform that offers users the convenience of watching videos and listening to music in one place.
- **MongoDB's Aggregation Pipelines** for 30% to 50% faster data processing and **Pagination** enabling 70% faster data fetching.
- Integrated **PayPal payment gateway** to enable secure transactions and premium upgrades.
- Implemented **JSON Web Tokens** for authentication and **OTP verification** for secure email validation.
- Built a responsive front-end with ReactJS, Redux Toolkit, and TailwindCSS for efficient state management and modern UI.

PROGRAMMING SKILLS

Data Structures

Arrays Strings Linked List Stack Queue Trees Heap Graph Tries

Algorithms

Searching Sorting Greedy Algorithms Recursion Backtracking Divide and Conquer Dynamic Programming Graph Algorithms

ACHIEVEMENTS

- Achieved **3rd Rank** in **DSA Coding Contest** with 350+ participants in **5th semester** organized by AKGEC, Ghaziabad.
- Achieved **4th Rank** in **DSA Coding Contest** with 300+ participants in **4th semester** organized by AKGEC, Ghaziabad.
- **Knight** at Leetcode [manas_agarwal16](#), **Pupil**, 1342 rating at Codeforces [manas_agarwal16](#), **3 Star** at Codechef [manasagarwal19](#)
- Solved **450+** Questions on **Data Structures and Algorithm** and **500+** Questions on **Competitive Programming**.

LEADERSHIP AND EXTRACURRICULAR

- Executed a series of **Workshops** focused on **Public Speaking Skills**, resulting in an increase in participant confidence levels by 40% as measured by pre and post-workshop surveys conducted among 50 attendees.
- Completed a **30-day challenge** focused on mastering **Backend Development** and documented progress on LinkedIn.